THE UNIVERSITY OF
NOTRE DAME
AUSTRALIA

MED 4000

WEEK 14

TUTOR GUIDE

Question 1

What are the clinical features of obstructive jaundice? Describe the differences between the LFTs in jaundice caused by obstruction vs hepatocellular damage.

Note that the results given below are not SI units. Please discuss the trends rather than the absolute numbers.

TABLE 46.12. Evaluation of Liver Function Tests in Jaundice.

Test	*Source*	*Pattern in gallstone obstruction of the biliary tree*	*Pattern in malignant obstruction of the biliary tree*	*Pattern in acute hepatitis*
Total bilirubin	Red blood cell destruction, hepatocyte processing	Elevated, but typically less than 10mg/Dl	Elevated, commonly more than 10mg/dL	Elevated
Direct bilirubin	Conjugation by hepatocytes	Elevated	Markedly elevated	Mildly elevated
Indirect bilirubin	Red blood cell turnover, hepatocyte processing	Minimally elevated	Elevated	Elevated
Alkaline phosphatase	Biliary epithelial cells and bone	Elevated	Markedly elevated	Minimally elevated
Transaminases	Hepatocytes	Minimally elevated	Minimally elevated	Markedly elevated
Gamma glutamyl transferase	Biliary epithelial cells	Elevated	Markedly elevated	Minimally elevated

BOX 18.2 JAUNDICE

- Jaundice is a yellowish discoloration of the tissues that becomes clinically apparent when serum bilirubin levels exceed 50 µmol/l (normal < 20 µmol/l)
- It may be due to excessive haemolysis, hepatic insufficiency or cholestasis; cholestatic (obstructive) jaundice is the type encountered in surgical practice
- **The two most common causes of surgical obstructive jaundice are cancer of the head of the pancreas and stones in the common bile duct (choledocholithiasis)**
- In cholestatic jaundice, the bilirubin has been conjugated by the hepatocytes and is therefore soluble in water and can be excreted in the urine; patients with obstructive jaundice typically have dark urine and pale stools and may have pruritus (thought to be due to the accumulation of bile salts)
- Obstructive jaundice is characterized by elevated serum alkaline phosphatase levels in addition to hyperbilirubinaemia, and may be accompanied by modest elevations in transaminase (aminotransferase) levels, reflecting liver damage

Question 2

What are the two main investigations used in patients with suspected pancreatic cancer?

The two main investigations are:

- **Endoscopic ultrasound (EUS): EUS is conducted by positioning the ultrasound against the pancreas to obtain futher information re solid and cystic lesions. This procedure also enables tissue to be taken via fine needle biopsy and cytological diagnosis.**
- **CT scan** is valuable in demonstrating the relationship of the tumour to the superior mesenteric vessels and portal vein. In addition, it may show lymphatic and hepatic metastases.

Other investigations include:
Liver function shows an obstructive pattern. The commonest difficulty is with intrahepatic cholestatic jaundice, where the pattern may overlap with extrahepatic obstruction.

Glucose tolerance is impaired in many patients.

Coagulation studies

In the presence of jaundice, prothrombin time is prolonged (see the International Normalised Ratio).

Imaging

Ultrasound excludes gallstones in the gallbladder and may show a normal common bile duct and demonstrate a dilated intra- and extrahepatic biliary tree, a mass in the head of the pancreas and possible liver metastases.

Endoscopy can demonstrate a malignant mass infiltrating the medial wall of the second part of the duodenum from which a biopsy may be obtained. **MRCP or ERCP** may indicate a malignant stricture of the common bile duct and often also the pancreatic duct ('double duct' sign). The stricture appears shouldered, abrupt and tight. Bile and pancreatic juice can be collected or ductal brushings or biopsy taken for cytological or histopathological analysis.

Percutaneous transhepatic cholangiography (PTC) is done by direct puncture of the bile duct through the liver substance, but in pancreatic cancer is a less useful investigation than an ERCP and is reserved for therapeutic biliary stenting when ERCP has failed.

Laparoscopy is used to exclude peritoneal or liver metastases before an attempt at resection.

Specific serum tumour markers

Both carcinoembryonic antigen (CEA) and carbohydrate antigen 19-9 (CA 19-9) are elevated in pancreatic cancer. CA 19-9 is 90% specific for pancreatic cancer but is excreted in bile, so estimations are inaccurate in patients with unrelieved jaundice. Although these markers are not usually at a high enough level to be diagnostic, they often support a clinical diagnosis, particularly if there is a progressive rise on repeated estimations.

Question 3

Cancer of the exocrine pancreas is an aggressive disease with a poor prognosis; the median survival from the time of diagnosis barely exceeds 5 months. Discuss the possible reasons.

Pancreatic cancer carries a poor prognosis, and most patients are dead within 2 years of diagnosis-more than half within 6 months. Even for those patients fortunate to present with a surgically resectable lesion, the 5-year survival after successful removal is less than 20%. **Possible factors contributing to poor prognosis:**

- **The main reason is due to the fact that less than 20% of pancreatic cancers are resectable** at the time of presentation, and these are predominantly lesions in the pancreatic head.
    - A carcinoma of the body or tail usually **presents late** because of the insidious progression of the tumour before symptoms occur. However, in retrospect there is often a non-specific prodromal phase of vague symptoms of malaise, weight loss and epigastric pain radiating to the back. Very few such patients have surgically resectable tumours by the time these symptoms occur. Exceptions to late diagnosis is often incidental body and tail tumours found during CT scanning.
    - The major clinical diagnostic problem is that **early symptoms mimic other commoner disorders** such as peptic ulcer, oesophagitis with heartburn, angina and biliary colic. Many will have spent several months undergoing investigation for these disorders before the correct diagnosis is established or jaundice develops. (If symptoms occur, the typical presenting history is of a middle-aged patient with obstructive jaundice and **pruritis**. There may be associated weight loss and epigastric pain which radiates through to the back and can sometimes be alleviated by sitting crouched forward.)
- 70% of tumours occur in the pancreatic head. **Spread of the growth** is by the four typical routes:
    - direct invasion of neighbouring tissues
    - lymph node involvement
    - blood-borne metastases to the liver and beyond
    - within the peritoneal cavity-transcoelomic spread.

- **Ineffective therapies**: Even with recent advances in surgery, anaesthesia, intensive care, cytotoxic chemotherapy and radiotherapy, there have been only modest improvements in outcome over the last fifty years. Chemotherapy regimens so far devised, either alone or in combination, are not curative.

Question 4

Henry underwent Whipple's operation and the surgical margins were clear. His postoperative course was uneventful and the jaundice subsided. Outline the objectives and methods of intervention in pancreatic cancer.

The objectives and methods of intervention are:

- resection of the primary lesion for cure (cure is achieved in <20% patients)
- alleviation of obstructive jaundice
- pain control
- treatment of exocrine failure and of diabetes
- chemotherapeutic palliation.

Surgical resection: Less than 20% of pancreatic cancers are resectable at the time of presentation, and these are predominantly lesions in the pancreatic head.

The modern one-stage operation is still often named a Whipple's operation after the originator of the previous two-stage procedure. Whipple first drained the biliary tree into the small intestine and secondly resected the tumour. Preliminary decompression of an obstructed biliary tree is now usually achieved by ERCP (see below). The operation is a major tour de force and should only be carried out in specialist centres by surgeons with extensive experience of pancreatic surgery. In such expert hands, the hospital mortality is less than 5%.

Tumours found at operation to be irresectable are palliatively treated by biliary and usually also duodenal bypass.

Alleviation of obstructive jaundice: Endoscopic, percutaneous radiological or surgical methods can be used to achieve this. Non-operative means are preferred for palliative relief of jaundice.

Chemotherapy regimens so far devised, either alone or in combination, are not curative. Regimens that have shown some palliative response (10-30%) are based on combination therapy using 5-fluorouracil (5-FU), adriamycin, epirubicin and mitomycin or methotrexate. Single agent gemcitabine is also widely used. Newer, less toxic derivatives of platinum-based compounds and metalloproteinase inhibitors are also now being evaluated.

Radiotherapy: Radioactive iridium wires can be inserted into the malignant mass at operation. Alternatively, intraoperative beam radiotherapy to the exposed tumour has been used. These methods have shown little additional benefit.

Question 5

Postoperatively Henry was considered a suitable candidate for a gemcitabine trial. However 2 cycles into the therapy he presented in emergency department with severe vomiting. His radiographs show small bowel obstruction. List the mechanical causes of bowel obstruction and discuss what is/are the most likely aetiology in this case.

Box 23.1 classification of mechanical obstruction

Luminal

- Gallstone (gallstone ileus)
- Food bolus

- Meconium ileus

Mural

- Stricture
  - o Congenital
  - o Inflammatory
  - o Ischaemic
  - o Neoplastic
  - o Intussusception

Extramural

- Adhesions
  - o Congenital
  - o Inflammatory
  - o Malignant
  - o Ischaemic
- Hernia
  - o External
  - o Internal
- Volvulus (twisting)
  - o Congenital
  - o Acquired

Box 23.2 **Clinical features of small-bowel obstruction**

- Colicky central abdominal pain
- Vomiting
- Abdominal distension
- Constipation

Abdominal pain is the first symptom and is central, ill-localised and characteristically colicky in nature, coming in waves with pain-free intervals of minutes. Large-bowel obstruction causes lower-abdominal colic and may eventually produce the more constant pain of distension. A strangulated loop in contact with the inner aspect of the abdominal wall causes well-localised, often severe, pain.

Vomiting follows the pain, and the higher the level of obstruction, the earlier and more profuse it is. Initially, upper-gastrointestinal contents are produced (food residue and dark greenish fluid) but, unless the obstruction is in the upper jejunum, dark brown, bitter, foul smelling (faeculent) material soon appears (vomiting of faeces is not a feature of intestinal obstruction but occurs in the relatively rare circumstance of an internal fistula between the large bowel and the stomach). In obstruction distal to the ileocaecal valve, vomiting may be absent because the small bowel can continue to propel its content into the distensible colon above the obstruction.

Distension is usually evident and is generally more marked the more distal the obstruction. Proximal jejunal obstruction (which is relatively uncommon) may be without distension.

Constipation. Most flatus passed per rectum is air that has been swallowed, so that complete obstruction at any level is associated with absolute constipation for both stool and gas. This occurs early in large-bowel obstruction and later in small-bowel obstruction. Incomplete obstructions, as may occur in the large bowel from a progressively encircling tumour, cause reduction in the size and frequency of bowel motions, with visible changes in the stool-mucus and blood.

Physical findings

Distension is present which is approximately related to the level of obstruction-the lower the obstruction in the small bowel, the greater the distension. In large-bowel obstruction, and because of the competence of the ileocaecal valve, the distension may outline the colon only, with a visible caecum. An abdominal scar suggests but does not prove that the cause of a small-bowel obstruction may be adhesions. Visible peristalsis may occasionally be seen in a thin patient and may coincide with an audible rush of peristalsis-see 'Auscultation' below.

Palpation may show:

- a mass anywhere in the abdomen-this suggests a specific cause for the obstruction
- an irreducible mass at a hernial orifice-a strangulated hernia
- tenderness and guarding-this is highly suggestive of strangulation but can be difficult to assess in the first few days after an abdominal operation.

Percussion produces a tympanitic note because of the presence of gas-filled loops of bowel, although in a slowly developing low small-bowel obstruction, fluid may predominate.

Auscultation reveals increased frequency of the segmenting sounds, which are high pitched and tinkling and have been likened to the sound of water lapping against the side of a small boat

Management

Non-operative approach can be used under the following circumstances:

- firm evidence that there is not a threat to the viability of the bowel-strangulation or perforation as suggested by signs of hypovolaemia, systemic inflammatory response and peritoneal irritation
- incomplete obstruction in either the small or large bowel with features which suggest non-progression, e.g. Crohn's disease in the small bowel
- some instances of complete small-bowel obstruction-the usual most suitable example is an adhesive obstruction (see below).

Conservative management involves:

- proximal decompression by a nasogastric tube with aspiration either continuously or on a regular intermittent basis
- water and electrolyte replacement
- repeated (4-6-hourly) evaluation of the clinical state-abdominal girth, development of tenderness, changes in bowel sounds and in cardiovascular status
- in individual circumstances, repeated plain X-rays or contrast studies and haematological and biochemical reassessment of the features of strangulation.

The indications for operating are:

- **established or suspected strangulation, including those with irreducible external hernia**
- **complete large-bowel obstruction with tenderness in the right iliac fossa-indicative of closed-loop obstruction with possible perforation of the caecum**
- **failure of resolution after a period of non-operative management,**
- **a cause (e.g. carcinoma) requiring surgical removal.**

ANSWERS

Question 1

Once considered a rare condition, adenocarcinoma of the oesophagus is now more common than SCC (of the oesophagus). What are the symptoms suggestive of oesophageal adenocarcinoma?

OESOPHAGEAL adenocarcinoma is predominantly a disease of overweight, middle-aged or older Caucasian men, with a female to male ratio of 1:9. By contrast, SCC is more common among people of Afro-Asian descent and affects females to males in a ratio of 3:4. Both cancers have a peak incidence between the ages of 50 and 60.

The early symptoms of both oesophageal and gastric cancers are vague and often difficult to define. Oesophageal lesions most commonly present with **progressive dysphagia**. Initially patients notice difficulty swallowing solids but, as the lesion progresses and occludes the lumen, even liquids pass with difficulty.

Pain on swallowing (odynophagia) is often the presenting symptom; constant pain suggests invasion into somatic structures.

Regurgitation and aspiration are common symptoms, especially when the patient is recumbent. These symptoms may be difficult to differentiate from the patient's usual reflux symptoms.

Hoarseness suggests spread to the recurrent laryngeal nerves, while coughing with swallowing may be indicative of a tracheo- oesophageal fistula.

Physical signs of oesophageal cancer are few. Supraclavicular lymphadenopathy, pleural effusions, hepatomegaly and bony tenderness may be an indication of distal disease.

Weight loss

Anaemia

Question 2

Describe the link between adenocarcinoma of the oesophagus and Barrett's oesophagus.

The risk of adenocarcinoma arising in Barrett's is said to be between one in 80 and one in 440 cases. This is around 30-40 times the risk of the general population. The risk of developing adenocarcinoma is approximately 0.5-1% per patient per year for patients with Barrett's, except for those with high grade dysplasia in which the rate of malignancy is much higher.

Adenocarcinoma arising from the true lower oesophagus generally arises in Barrett's oesophagus. Barrett's oesophagus is defined as the replacement of the squamous epithelium in the lower oesophagus by a columnar epithelium, with histological findings of intestinal metaplasia. Over time, there is the risk of dysplasia occurring within the metaplastic region, which can ultimately progress to adenocarcinoma.

While gastro-oesophageal reflux is a clear risk factor for Barrett's oesophagus, the risk factors for adenocarcinoma are less well defined. Obesity and cigarette smoking and perhaps alcohol have all been shown to increase the risk of malignancy.

Question 3
Only 20-30% of those with oesophageal and oesophageal-gastric junction (OGJ) tumours are considered suitable for curative treatments. Outline the surgical approaches used for resection of the oesophageal tumour.

MANAGEMENT strategies for patients with oesophageal cancer may have either a curative or palliative intent. Cases may be discussed in a multidisciplinary setting, ideally with a pathologist, radiologist, surgeon, medical oncologist, radiation oncologist, cancer support nurse and dietitian.

After patients have been staged and assessed for fitness, only 20- 30% of those with oesophageal and oesophageal-gastric junction (OGJ) tumours are considered suitable for curative treatments.

Oesophagectomy remains the mainstay of curative treatment for OGJ and oesophageal lesions. Regardless of the approach to oesophagectomy it is a morbid procedure, with mortality rates of up to 5% in even the most specialised centres. **Five-year survival rates after surgery are generally only 20-40% overall.** There is a range of surgical approaches for resection of the oesophageal tumour. Choice depends on a number of factors including the

- site of the tumour
- general health of the patient
- experience and preference of the surgical team.

Abdominal approach: The incidence of cancers at the oesophago-gastric junction (OGJ) is increasing, although the reasons for this are presently not known. The standard operation for a proximal gastric cancer is a radical total gastrectomy, although the proximal gastrectomy does have its proponents. If thelesion encroaches onto the oesophagus, then the issue arises of getting an adequate oesophageal margin. The higher the point of division, the more technically difficult the anastomosis.

Abdominal and right chest (Ivor Lewis): This is the standard approach for most oesophago-gastric surgeons. It is suitable for lesions of the mid and distal oesophagus. The first phase of surgery is the laparotomy, at which the stomach and duodenum are mobilised and a pyloroplasty and (often) a feeding jejunostomy performed. The abdominal wound is then closed and the patient repositioned for a posterolateral thoracotomy. The thoracic oesophagus is then mobilised and the stomach/gastric tube delivered into the chest. The specimen is resected and the anastomosis performed.

Subtotal (three stage, McKeown): This is essentially the same approach as the abdominal/ right chest except the right chest is closed after the entire thoracic oesophagus has been mobilised. The patient is then repositioned supine whereupon the neck is opened (usually the left side) and the specimen delivered through the neck for resection and anastomosis. Advantages of this approach are a slightly larger extent of resection, and the fact that the anastomosis is in the neck rather than the chest. The disadvantage is the increased operating time, the incidence of postoperative swallowing difficulties, and the need to heal three incisions.

Trans-hiatal oesophagectomy (Orringer): This technique was developed to avoid the need for a thoracotomy, especially in patients with marginal respiratory status. It involves synchronous surgery from the abdomen and the neck, with the oesophageal hiatus being widely opened from the abdomen and the superior mediastinum being approached through the neck. A formal nodal dissection is not possible. Even in the routine case, blood loss is heavier, and there is obviously the risk of catastrophic bleeding from the mediastinum. While the procedure does have its proponents, many oesophago-gastric surgeons would argue that radiation or chemo-radiation is most appropriate for otherwise curable patients who will not withstand a thoracotomy.

Minimally invasive oesophagectomy is increasingly widely performed. The operation may involve thoracoscopic, laparoscopic or combined thoracoscopic and laparoscopic resection. A left cervical anastomosis is usually made.

Question 4

What are the approaches used in palliation of severe dysphagia in a patient with oesophageal cancer?

Palliation of dysphagia by mechanical means

If a patient is seen before the development of severe dysphagia, it is often possible to control this symptom with radiation, chemotherapy alone (less commonly), or chemo-radiation. However, if the patient presents with severe dysphagia, then a mechanical solution is necessary to overcome the obstruction.

Serial endoscopic dilatation: This is of historical interest only as the reported rates of perforation are very high at 5–10% (remembering that many of those perforations occurred in patients with very advanced lesions that were not being treated in any other way).

Laser ablation: The NdYAG laser is widely used in this setting (in selected lesions). The best lesions are non-circumferential and form a ridge that projects into the oesophagus. Laser is probably preferable to a metal stent in this group. The NdYAG laser delivers maximal energy 2–3 mm deep to the surface and therefore perforation is a risk, especially in stenosing lesions where the direction of the lumen is not obvious.

Argon plasma coagulation: This is a form of monopolar diathermy that conducts the energy to the tissues in a superheated 'plasma' of argon. Tissue damage is much more superficial than with laser, and the risk of perforation is lower Indications are similar to laser and early results seem comparable.

Self expanding metal stents: These are the mainstay of the management of malignant dysphagia. Early complication rates are low (2% mortality) and 95% of patients have significant improvement in their dysphagia. Stent migration occurs in only 5% and obstructive episodes requiring intervention in 3% of patients. The great advantage over the older plastic stents are the smaller delivery systems which make them safer to deploy, and the greater internal diameter once fully expanded.

ANSWERS

Question 1

What can you tell Ewan about his risk of developing colorectal cancer with one relative confirmed with CRC? If the overseas aunt was a sister of Martin's, how would that affect his risk level?

Table 1: Relative risk of colorectal cancer with different family histories, and NHMRC/Australian Cancer Network guidelines on screening*

Family history of CRC	Relative risk	Recommendations
No family history, normal population risk	1 (population risk)	FOBT biannually from age 50
One first degree relative with CRC diagnosed at after age 55	Up to a twofold increase	FOBT biannually from age 50. Consider sigmoidoscopy every five years
One first-degree relative with CRC diagnosed before age 55 or two first- or second-degree relatives on one side of family diagnosed at any age	3-6-fold increase	Five-yearly colonoscopy from age 50. Consider FOBT annually in intervening years

*Note: These guidelines apply to individuals who do not have a known genetic predisposition to CRC, such as HNPCC or the polyposis syndromes.
CRC = colorectal cancer; FOBT = faecal occult blood testing; HNPCC = hereditary non-polyposis colorectal cancer

Question 2

Ewan asked about genetic screening. Describe briefly a) what you would say to Ewan about familial colorectal cancer and genetic screening in general and b) what information you would provide about Hereditary Cancer Registers (HCR) which have links to Familial Cancer Clinics.

a) Familial colorectal cancer

Between 2–5% of patients with colorectal cancer (CRC) have inherited a known genetic mutation. The types of CRC known to involve genetic susceptibility are:

- **familial adenomatous polyposis (FAP)** - FAP is a rare condition, usually due to a mutation in the adenomatous polyposis coli (APC) tumour suppressor gene. Without treatment, those with proven FAP have a lifetime risk of CRC of almost 100%. Individuals with a mutated APC gene usually develop hundreds of adenomas throughout the colon and rectum that may appear as early as the teenage years. If left untreated, one or more of these adenomas will progress to cancer, often at an early age.
- hereditary nonpolyposis colorectal cancer (HNPCC/Lynch syndrome).

Predictive genetic testing to determine risk in unaffected members of high risk families requires the identification of the family specific genetic mutation in an affected relative. **Mutation searching for familial cancer mutations is an expensive and often lengthy process that can potentially produce 'uninformative' results.**

- Those found to HAVE inherited a gene mutation that confers a high risk of developing cancer can be offered individualised cancer screening and strategies for prevention.

- Relatives proven NOT to have inherited the family specific mutation still have an average risk of developing the cancer based on their age and should follow recommendations for general population screening. However, they can be spared the intensive screening needed by someone who has/may have the mutation.

b) The Hereditary Cancer Registers (HCR) assists with the management of individuals from families at high risk for certain types of hereditary cancers. There is an advisory committee of consumers and specialist clinicians. Conditions covered by the registers are:

- hereditary nonpolyposis colorectal cancer syndrome
- familial adenomatous polyposis (FAP)
- Peutz-Jeghers syndrome
- juvenile polyposis, and
- other polyposis syndromes.

Familial Cancer Services exist in most Australian states and territories and provide risk assessment, genetic counselling and, if appropriate, genetic testing for a causative mutation. The service usually confirms cancer diagnoses in relatives to account for the uncertainty in people's self reported family history.

Table 1. Hereditary cancer registers in Australia

State	Conditions covered	Address	Telephone
New South Wales/ Australian Capital Territory	FAP, HNPCC, polyposis syndromes, VHL	Locked Bag 9000 Potts Point NSW 1335	02 9334 1807 02 9334 1794
Queensland	FAP, polyposis syndromes	Qld Clinical Genetics Services Level 4, Block 65 Royal Children's Hospital Herston Qld 4029	07 3636 5117
South Australia	FAP, HNPCC, polyposis syndromes, other familial cancers	Familial Cancer Unit (FCU) Women's and Children's Hospital 72 King William Road North Adelaide SA 5006	08 8161 6995
Victoria	FAP, HNPCC, breast, breast/ovary	The Cancer Council Victoria 1 Rathdowne Street Carlton Vic 3053	03 9635 5176 03 9635 5374 03 9635 5414
Western Australia	FAP, HNPCC, polyposis syndromes, breast/ovary	Genetic Services of WA 374 Bagot Road Subiaco WA 6008	08 9340 1603 08 9340 1713
Tasmania	FAP, HNPCC, polyposis syndromes	PO Box 1963 Launceston Tas 7250	03 6348 7006

Question 3
What is the difference in screening approaches for colorectal cancer in the general population compared to that of patients at increased risk, like Ewan?

If a patient is found to have a colorectal adenoma, or multiple adenomas, their lifetime risk of colorectal cancer is significantly increased even after removal of the polyp. The current NHMRC guidelines recommend these patients have a repeat colonoscopy after 4-6 years.

a) General population screening for colorectal cancer has only just begun since August 2006, and in a very limited fashion; only 50-, 55- and 65-year-olds are being screened, not biannually from 50 as per clinical trial evidence and NHMRC guidelines. If one looks purely at the evidence from clinical trials of the benefits of population screening in terms of life years saved per dollar spent, faecal occult blood testing (FOBT) screening for colorectal cancer is only slightly less cost effective than mammographic screening for breast cancer, and considerably more effective than Pap smear screening for cervical cancer, while the effects of prostate cancer screening remain unproven altogether.

Other forms of population screening:
The only other screening modality evaluated to any degree for colorectal cancer is flexible **sigmoidoscopy** with no bowel preparation and no sedation. An oft-quoted retrospective case–control study showed a significantly lower likelihood of colorectal cancer developing in patients who have had a flexible sigmoidoscopy. On examining this study, the flexible sigmoidoscopy group had few significant polyps removed. Therefore unless one believes that performing a flexible sigmoidoscopy itself is in some way protective, the result is almost certainly the result of selection bias.
The use of **colonoscopy** (examining the entire large bowel after bowel preparation and with sedation) has not been evaluated for population screening, nor is it a viable option because of resource limitations.
'Virtual colonoscopy' (helical CT with computer generated 3D reconstructions of the colon) has been suggested as a possible screening modality. It may become an option in the future. However, at present the resolution and accuracy are not sufficient, nor has it been properly evaluated for this indication.

The carcino-embryonic antigen (CEA) test is occasionally incorrectly used as a screening test in patients without colorectal cancer. CEA is not elevated in people with polyps (the targets of screening), often not elevated in early colorectal cancer, and usually signifies advanced or metastatic disease when elevated. It is also frequently elevated in heavy smokers, other benign and malignant conditions and occasionally even in normal individuals, with these false positives leading to anxiety and unnecessary investigations. It is therefore a poor screening test.

b) Screening of patients at increased risk: Another crucial form of screening for colorectal cancer over and above general population screening is **more intensive screening of higher-risk individuals, namely those with a family history of colorectal cancer.** With general population screening, compliance is often low (20-30%) and those who do comply are often the least at risk (the 'worried well'). By identifying higher-risk individuals and implementing appropriate targeted screening programs, GPs can make a major impact by directing limited resources to those most likely to be at risk. While all patients with a family history of colorectal cancer have an increased risk of developing colorectal cancer, the risk can be further quantified. This is outlined in the NHMRC/Australian Cancer Network guidelines, and screening recommendations for different categories of risk conferred by different family histories are also outlined (see table 1).

MOST cases of colorectal cancer start out as benign adenomatous polyps, which are asymptomatic and grow for many years before becoming malignant. This polyp-to-cancer sequence takes 5-10 years. Therefore there is a latent period when patients have benign colonic polyps that can be removed, and colorectal cancer prevented. Studies have shown that screening programs in which adenomas are removed significantly lower the risk of developing colorectal cancer.
Colorectal cancer has a latent pre-malignant phase, and has a well-understood natural history, with evidence that it can be modified by a cost-effective screening program. That is, it fulfils all the criteria for population screening.

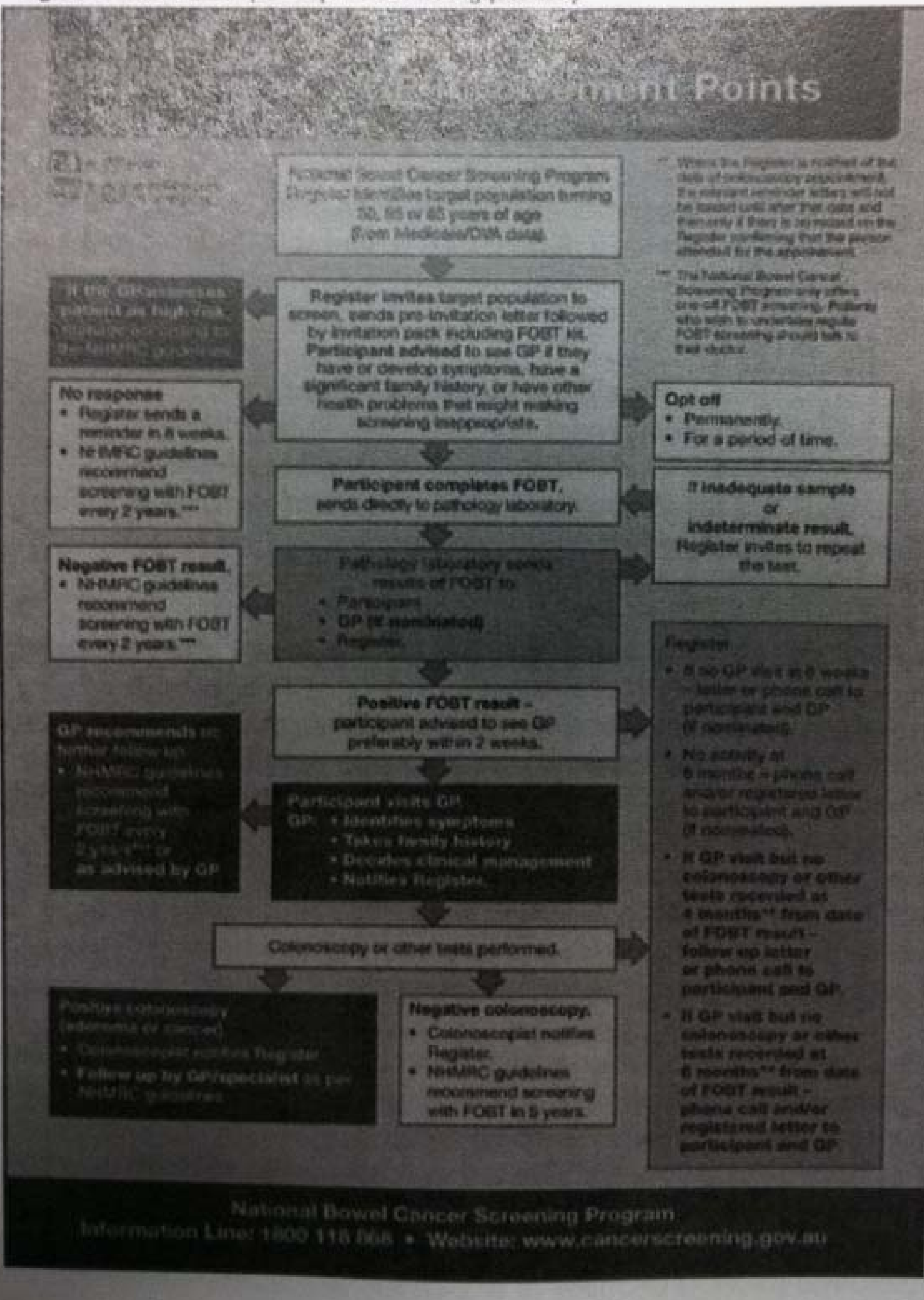
Question 4

Review the National Bowel Cancer Screening Program [3_7_2] and summarise the participant screening pathway.

National Bowel Cancer Screening Program

The NBCSP pathway is summarised in *Figure 1* below. Individuals in the eligible age ranges are identified through Medicare or Department of Veterans' Affairs data and mailed a FOBT kit in the post. The NBCSP register maintains information about participant progression through the screening pathway.

Figure 1. The NBCSP participant screening pathway



ANSWERS

Question 1

Secondary liver tumours are 20 times more common than primary ones. Which are the common sites of the primary tumours?

In 50% of cases, the primary tumour is in the **gastrointestinal tract**; other common sites are the **breast, ovaries, bronchus** and **kidney**. Almost 90% of patients with hepatic metastases have tumour deposits in other sites.

Question 2

Applying your knowledge of regional anatomy, which structures should not be involved for the surgeon to consider the possibility of surgical resection?

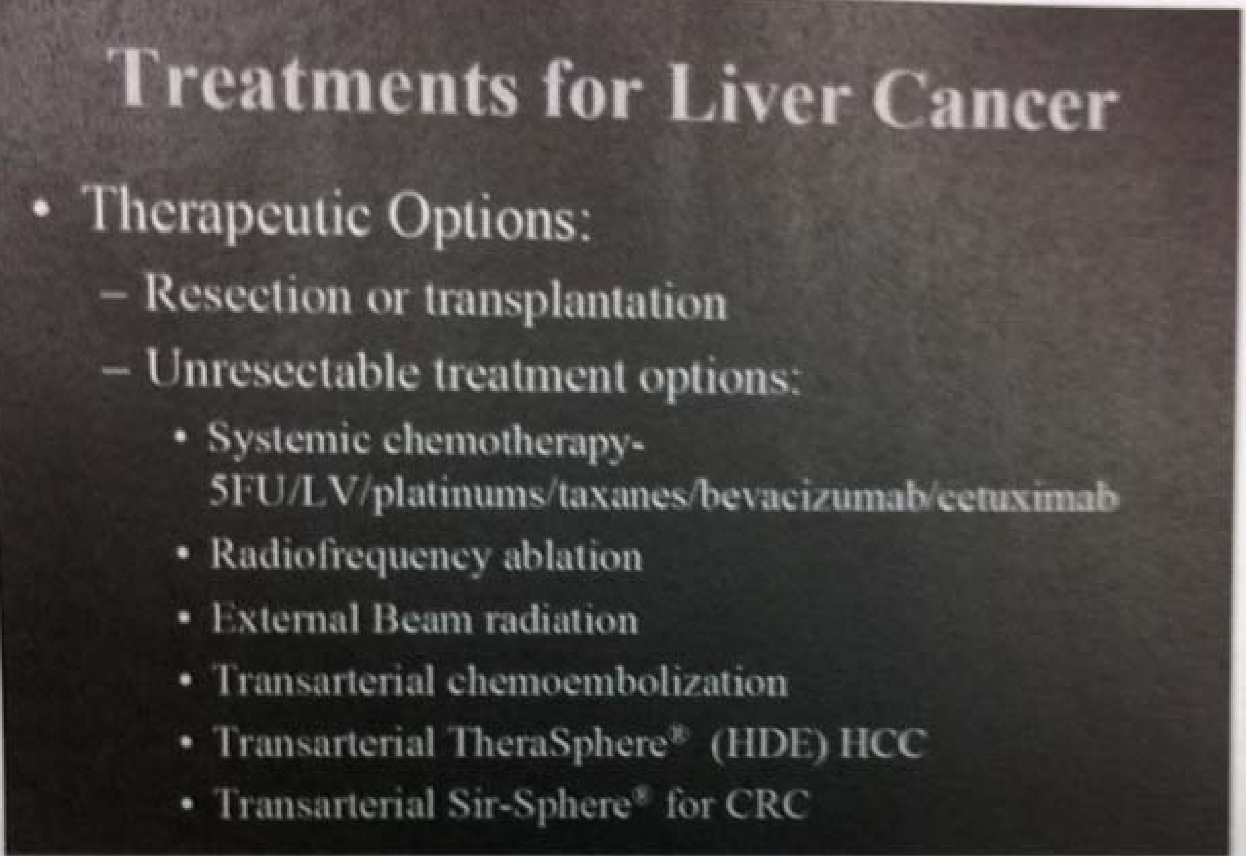
CCO: Treatment of Resectable Metastatic Disease of the Liver

Metastatic liver disease may be treated with resection so long as there is no radiographic evidence of involvement of the hepatic artery, major bile ducts, main portal vein, or celiac/para-aortic lymph nodes.[Adam 2008] In addition, patients must be predicted to retain adequate hepatic function following resection. For this approach to be efficacious and potentially curative, complete resection of metastatic lesions must be feasible, there should be no unresectable extrahepatic lesions, and the primary tumor should also be resected, either in the past or during resection of metastases.[NCCN 2010a]

A potential advantage of neoadjuvant therapy involves downsizing the primary tumor and metastatic lesions prior to surgery. Patients with unresectable disease were treated with chemotherapy. A favorable response to neoadjuvant chemotherapy was seen in 138 patients (12.5%) who then went on to receive secondary hepatic resection. The National Comprehensive Cancer Network guidelines suggest that neoadjuvant therapy with leucovorin, 5-fluorouracil, and irinotecan (FOLFIRI), FOLFOX, or XELOX with or without bevacizumab or FOLFOX, or FOLFIRI with or without cetuximab or panitumumab (in wild-type *KRAS* tumors only) may be of benefit for resectable liver or lung metastases. The most important aspect of employing resection in colorectal cancer liver metastases is to include a multidisciplinary team of experts for decision-making to identify the optimal treatment strategy.

Question 3

When resection is not possible nor indicated, which non-surgical liver-directed therapies are currently available for patients with liver metastases?



When resection is not possible or not indicated, nonsurgical liver-directed therapies are given to treat hepatic metastasis, with treatment goals being alleviation of symptoms and prolongation of survival.

Transarterial chemoembolization (TACE) has been used to treat hepatic metastases with response rates as high as 70% to 90%. In a single-institution study of 28 patients, 68% experienced improvement in their symptoms, and 69% had objective radiographic response following transarterial chemoembolization. Improvement in techniques has reduced the occurrence of complications, specifically hepatic necrosis, renal failure, and sepsis.

Radiofrequency ablation has also been used to treat liver metastases, although less often than transarterial chemoembolization. Candidate tumors are generally smaller than 5 cm and fewer than 4 in number. Small case series have reported low rates of treatment failure, and 80% patients of patients experience alleviation of symptoms. The adverse effects of radiofrequency ablation are fewer than with transarterial chemoembolization.

Focal radiation therapy using 90 Yttrium has also been employed in unresectable tumors with success.

Question 4

Which anatomical feature (blood supply to the liver) has been used in the adoption of localised radiation therapy of liver metastases with yittrium-90 using an innovative delivery mechanism ?

Pathophysiology

Normal Liver

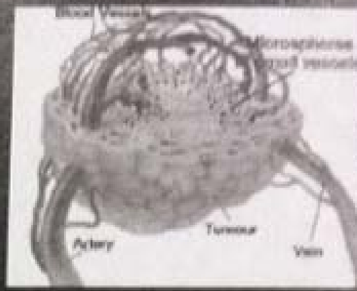
- 75% of blood supplied by P.V.
- 25% of blood supplied by H.A.

Hepatic Tumors

- Majority of the blood supply from H.A.
- Hypervascular relative to normal liver tissue

Role of Y-90 in Liver tumor

- Y-90 is introduced via catheterization of the hepatic artery
- Microspheres are trapped in the tumor capillary bed where they exert a local radio-therapeutic effect.



Yttrium-90 Radioisotope

- 100% pure beta emitter, 0.9367 MeV
- Physical half-life of 64.2 h
- Irradiates tissue with average range of 2.5 mm
- Maximum penetration of approx. 1.0 cm
- A constituent of an insoluble glass/resin
- Microsphere diameter 25µ - 35µ

A viable treatment option for inoperable liver cancer

SIR-Spheres™

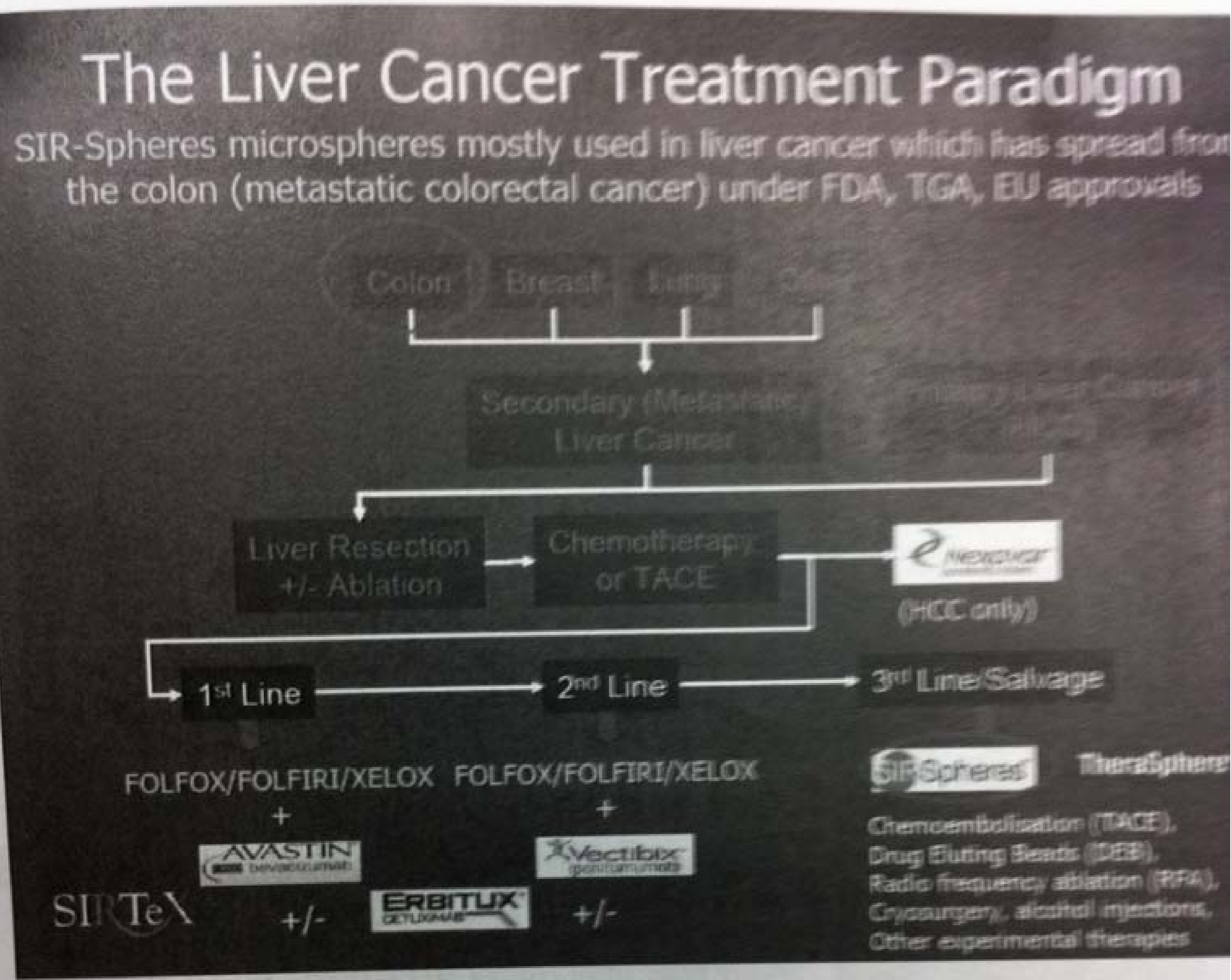
Regulatory Approval Status

Australia (TGA) 1997
USA (FDA) 2002
EU (CE Mark) 2002

Selective Internal Radiation Therapy (SIRT)

To selectively target a very high radiation (Yttrium-90) dose to all tumours within the liver, regardless of their cell of origin, number, size or location while at the same time maintaining a low radiation dose to the normal liver tissue

SIRTeX



The Liver Cancer Treatment Paradigm
SIR-Spheres microspheres mostly used in liver cancer which has spread from the colon (metastatic colorectal cancer) under FDA, TGA, EU approvals
Colon
Breast
Liver Resection +/- Ablation
Chemotherapy or TACE
(HCC only)
1st Line
2nd Line
3rd Line/Salvage
FOLFOX/FOLFIRI/XELOX
+
AVASTIN
FOLFOX/FOLFIRI/XELOX
+
Vectibix
SIRTeX
+/-
ERBITUX
+/-
SIR-Spheres
TheraSphere
Chemoembolisation (TACE),
Drug Eluting Beads (DEB),
Radio frequency ablation (RFA),
Cryosurgery, alcohol injections,
Other experimental therapies

ANSWERS

Question 1

What is the differential diagnosis in this patient?

Table 1: Differential diagnoses for acute scrotal pain

Condition	Most common age	Pain history	Risk factors	Examination	Bloods/urinalysis	Ultrasound
Testicular torsion	1. 12-30 years 2. Neonates	Severe sudden-onset testicular pain ± nausea	'Bell-clapper' deformity Cryptorchidism Trauma	High-riding/horizontal testis Unilateral absence of cremasteric reflex Globally tender testis	Elevated WCC (60%) Usually normal urinalysis	Reduced or absent blood flow to affected testis
Torsion of appendix testis	Prepubertal	Gradual onset of testicular pain	Prepubertal age	Tender superior pole of testis 'Blue dot sign' visible on affected side of scrotum	Usually normal urinalysis	Hypoechoic nodule Normal testicular blood flow
Epididymitis/orchitis	Postpubertal	Gradual onset of moderate testicular pain ± dysuria	Sexually active Genitourinary instrumentation Previous UTIs/STIs Chronic retention	Tender epididymis Scrotal swelling and erythema — 'orange peel' appearance 'Prehn's sign' — elevating the scrotum decreases pain	Pyuria Bacteriuria	Epididymal and/or testicular swelling Increased blood flow to testis Abscess or reactive hydrocoele
Tumour	30-40 years	Variable: painless, dull ache or post-traumatic pain	Cryptorchidism Contralateral testicular tumour First-degree family history	Firm testicular mass Gynaecomastia (in 30% of patients with Leydig cell tumours) Evidence of distal disease — hepatomegaly, bone pain	Elevated tumour markers (AFP, beta-hCG, LDH)	Testicular mass ± reactive hydrocoele
Trauma	Any age but usually <35 years	Moderate or severe scrotal pain after injury	High-impact sports Other blunt trauma	Swollen tender scrotum Poorly defined testis	Normal or haematuria	Evidence of haematocoele or testicular rupture

WCC = white cell count, AFP = alpha fetoprotein, beta-hCG = beta human chorionic gonadotrophin, LDH = lactate dehydrogenase

Question 2

In Emergency Department, he was sweaty and pale and in severe pain. After cutting off his shorts, the ED doctor noted that the left side of the scrotum was swollen and extremely tender and positioned at a higher level than the right side. Moreover the cremasteric reflex on the left side was absent. Provisional diagnosis of testicular torsion was made and Carlos was immediately prepared for surgical exploration.

What is testicular torsion and why is it considered a true surgical emergency that must be differentiated from the other causes of unilateral testicular pain?

Testicular torsion is the rotation of the testis on the spermatic cord, resulting in venous congestion, testicular ischaemia and, later, necrosis. It can occur at any age but predominantly affects males aged 12-30, with a peak at 14. There is also a smaller peak occurring during infancy. Testicular torsion accounts for 16-42% of all presentations of acute scrotum. The left testis is more commonly affected and 2% of cases are bilateral.

Most testicular torsion results from an anatomical anomaly found in 12% of asymptomatic men — the "bell-clapper" deformity, when the tunica vaginalis surrounds the entire testis and epididymis, resulting in deficient posterior fixation of the testis. The testis is then free to rotate within the tunica vaginalis like a gong (clapper) in a bell, predisposing to intravaginal torsion.

Testicular torsion may be precipitated by exercise, sexual activity and trauma, but often occurs at rest. In the patient with severe testicular pain persisting for longer than one hour after trauma, testicular injury or torsion should be excluded.

The two most important factors determining testicular salvage after torsion are

- **time to detorsion and**
- **degree of testicular rotation.**

When detorsion is undertaken within 6 hours of symptom onset, testicular viability approaches 100%, decreasing to 20% after 12 hours and virtually 0% after 24 hours.

The 3 most important clinical predictors of testicular torsion are:

- Onset of pain <6 hours.
- Absence of the cremasteric reflex.
- Diffuse testicular tenderness.

Question 3

Describe the key features in history and examination that would indicate testicular torsion. How is the diagnosis confirmed and what is the management?

Clinical features

The patient with testicular torsion characteristically describes a sudden onset of severe unilateral scrotal pain. Associated nausea and vomiting is common, as is abdominal pain. Testicular pain radiating into the groin and lower abdomen may be mistaken for abdominal pathology unless the genitalia are carefully examined.

Examination

Scrotum: The overlying scrotum may be swollen and erythematous in both torsion and inflammatory conditions.

Testis: The torted testis is usually swollen, globally tender to palpation and positioned higher and more horizontal than the contralateral testis, because of twisting of the spermatic cord. The last of these findings is highly suggestive of torsion.

Spermatic cord: The torted spermatic cord may be palpable as a thickening above the affected testis. Normal position of the epididymis does not exclude torsion, as the testis may have rotated 360° or 720°. Elevation of the scrotum may improve the pain of epididymitis (Prehn's sign) but not the pain of torsion.

Cremasteric reflex: Normally a positive cremasteric reflex is found in most males over the age of 30 months and is defined as a minimal 0.5cm elevation of the ipsilateral scrotum on pinching or stroking of the upper medial thigh. In the patient with acute scrotal pain ipsilateral loss of the cremasteric reflex is the most sensitive physical finding for diagnosing testicular torsion.

Investigations

If testicular torsion is suspected clinically, imaging should not delay prompt surgical referral for exploration.

White cell count may be raised in up to 60% of cases of testicular torsion, and is unreliable for differentiating torsion from epididymitis.

Urinalysis should be performed in all patients presenting with acute scrotum, and is usually normal in torsion.

Because of the delay involved, imaging should only be performed in patients with equivocal clinical features and a low suspicion for torsion.
Colour Doppler ultrasonography is the imaging modality of choice, with a sensitivity of 88% and specificity of 100% for detecting testicular torsion. In addition the presence of tumour or extra-testicular pathology such as varicocoele, hydrocoele or haematoma can be visualised.
Doppler ultrasound usually shows reduced or absent intratesticular blood flow of the torted testis, compared with the contralateral side, while testicular blood flow is increased in inflammatory conditions.
A nuclear medicine scan is highly sensitive for testicular torsion. However, its use is limited by poor access and delays associated with obtaining results.

Management
When testicular torsion is suspected, immediate surgical scrotal exploration is the management of choice. If the testis appears viable after detorsion, orchidopexy is performed. Forty per cent of men with the bell-clapper anomaly have a similar deformity on the contralateral testis, so orchidopexy is usually performed bilaterally.

If operating facilities are not immediately available, manual detorsion may be attempted. Testes usually tort towards the midline, therefore the un-torting action has been likened to opening the pages of a book, with the affected testis rotated outwards. These patients still require radiological and surgical assessment to confirm detorsion, as a testis can tort up to 720°, and manual detorsion may have only partially relieved the torsion.

Prognosis
Despite detorsion, up to 40% of patients with unilateral torsion have subsequent subfertility, likely due to disruption of the blood-testis barrier. Sperm counts and degree of testicular atrophy relate to duration of torsion. Note that a missed torsion can later present as atrophy of the affected testis.

Question 4
How do patients with testicular tumours present? What investigative procedure is most useful in differentiating testicular tumours from other causes of testicular masses?

Typically patients present with a unilateral painless lump detected by themselves or their partner. Some report regional pain due to haemorrhage or infarction within the tumour, torsion or incidental preceding trauma. The most common misdiagnosis at initial presentation for this group is epididymitis.
In 10% initial presentation is with symptoms of systemic metastases. This includes:
- Neck lumps from supraclavicular nodal metastases.
- Cough or dyspnoea from pulmonary metastases.
- Abdominal symptoms or bone pain from retroperitoneal disease or bony involvement.
- Lower-limb oedema from iliac venous compression or thrombosis.

The systemic endocrine effects of the tumour may result in gynaecomastia (5% with testicular germ cell tumours) or precocious puberty in younger patients.
On examination each testis should be careful palpated, taking note of any diffuse rubbery

enlargement or focal, hard irregular areas. The normal testis can be used as a reference. Each hemi-scrotum can then be examined for a reactive hydrocoele and local tumour involvement, followed by thorough systemic examination for evidence of distant metastases.

The incidence of testicular cancer is second only to melanoma in men aged 18-35. Its incidence h doubled over the past 30 years in most industrialised nations, but survival rates even for metastatic disease remains excellent. The mean age of presentation is 30, with Australian males having a one in 221 lifetime risk of developing testicular cancer before age 75.

Risk factors for testicular tumours include cryptorchidism, family history of testicular malignancy first degree relatives, tumour of the contralateral testis and Klinefelter's syndrome.

Most testicular cancers (95%) are germ cell tumours; the remainder are either gonadal stromal tumours (5%) or metastases from another site (<1%). Testicular germ cell tumours are divided into two broad histological categories for treatment purposes: pure seminoma and non-seminomatous germ cell tumours. Seminomas and mixed germ cell tumours are most common in postpubertal males up to age 40.

Any patient with a palpable testicular mass is presumed to have testicular cancer until proven otherwise.
Ultrasonography is the preferred investigation and can distinguish between orchitis, intratesticular tumours and extratesticular masses such as spermatocoele or hydrocoele. Most germ cell tumours are evident as well defined heterogeneous hypoechoic intratesticular lesions. There may be focal necrosis or calcification.

Males with inguinoscrotal symptoms and an unclear diagnosis, or in whom testicular examination is limited by a large hydrocoele or varicocoele also warrant diagnostic ultrasonography. In patients with suspected testicular cancer on examination and ultrasonography, further investigations and urological referral should be expedited.

Question 5
Discuss the reasons for delay in diagnosis in patients with testicular lesions.

The diagnosis of a testicular malignancy can be delayed by both patient and physician factors. Patients may misunderstand the significance of their symptoms or simply ignore a scrotal lump for fear of the diagnosis. Physicians too may delay diagnosis by attributing symptoms to benign conditions such as epididymitis, hydrocoele or varicocoele. A diagnosis may be missed by omitting testicular examination in patients with abdominal symptoms, or by failing to image patients with suboptimal testicular examination.

Promoting regular testicular self-examination, and maintaining vigilance for testicular cancer in patients presenting with inguinoscrotal symptoms aids prompt diagnosis and management.

Question 1

Describe the common clinical presentations of renal cell carcinoma.

Key messages

- Incidental renal masses are now commonly found in the course of unrelated imaging investigations.
- Simple renal cysts are common, increasingly so beyond age 50; if asymptomatic they require no further investigation.
- Complex renal cysts that enhance with contrast on CT must be regarded as malignant.
- Up to 30% of solid renal masses <4cm in diameter may be benign, but the role of renal biopsy remains controversial in this setting. Nephron-sparing approaches such as laparoscopic partial nephrectomy will become more widely used in these cases.
- Laparoscopic radical nephrectomy is an accepted treatment for T1 renal cell carcinoma.
- Nephrectomy with immunotherapy for metastatic disease is currently practised in many units. Other chemotherapeutic approaches to renal cell carcinoma are under investigation.

RCC accounts for about 3% of all adult malignancies and its incidence is increasing, largely due to increased detection through the use of ultrasound and CT imaging techniques. It has a male to female predominance of 3:2, and the mean age at diagnosis is 70 years. About **one-half of RCCs are now diagnosed incidentally,** and about one-third of all RCC cases have overt metastases at diagnosis. The prognosis of asymptomatic RCC is better than for RCC that presents with local symptoms (such as pain and haematuria), which in turn is better than that which presents with systemic symptoms.

More than 50% of RCCs are found incidentally. The triad of loin pain, haematuria and palpable abdominal mass is now rare. Twenty per cent present with *paraneoplastic syndromes*. RCCs can produce 1,25 dihydroxycholecalciferol, renin, erythropoietin and various prostaglandins in pathological amounts. Consequently hypercalcaemia, hypertension, polycythaemia, anaemia, cachexia, pyrexia and raised ESR are well described in association with RCC. These generally resolve after nephrectomy in suitable cases. In advanced cases examination may reveal palpable cervical lymphadenopathy, or varicocoele and lower limb oedema, which may suggest venous involvement.

Although RCC invades locally, it has a tendency to invade venous structures, seen in 10% of RCCs. This is usually seen as tumour thrombus on imaging, which can extend up the inferior vena cava as far as the right atrium.

Diagnosis and the role of renal biopsy: The diagnosis of RCC has traditionally only been radiological, as the rates of malignancy within renal masses were very high. The indication for fine-needle aspiration biopsy (FNAB) was generally reserved for specific situations. FNAB should also be considered in patients with a renal mass and a history of primary non-renal malignancy, when a renal metastatic lesion is a possibility. However, in most cases of RCC the diagnostic value of biopsy is said to be limited by difficulties in sampling and interpretation, such that it adds nothing to diagnosis by imaging alone.

Question 2

What are paraneoplastic syndromes? Describe the underlying pathophysiological mechanisms.

araneoplastic syndrome: The collective signs and symptoms caused by a substance emanating rom a tumor or in reaction to a tumor. Paraneoplastic syndromes can be due to a number of auses including hormones or other biologically active products made by the tumor, blockade of he effect of a hormone, autoimmunity, immune-complex production, and immune suppression. or example, a well-known paraneoplastic syndrome involves inappropriate ADH secretion. ADH is ntidiuretic hormone. The syndrome disturbs fluid balance, results in the inability to put out dilute rine, and causes nausea, vomiting, muscle cramps, confusion, and convulsions. Inappropriate DH secretion can occur, for example, with oat-cell lung cancer, pancreatic cancer, prostate ancer, and Hodgkin disease.

A paraneoplastic syndrome is not produced by the primary tumor itself or by its metastases, nor is t caused by compression, infection, nutritional deficiency, or treatment of the tumor.

Paraneoplastic syndromes in renal cell carcinoma:

Event	Cause
Raised ESR	Changes in plasma proteins
Anaemia	Depressed erythropoiesis and haemolysis
Polycythaemia	Erythropoietin secretion
Hypercalcaemia	Tumour secretion of parathormone-like substance
Raised alkaline phosphatase concentration	Secretion from the tumour
Pyrexia	Circulating pyrogens
Hypertension	Secretion of renin
Amyloid deposition	Unknown
Peripheral neuropathy and myopathy	Unknown

ANSWERS

Question 1

Describe the agents that are thought to be associated with bladder cancer development.

Table 1: Influential agents in bladder cancer development		
Risk factor	**Carcinogenic influence**	**Risk**
Cigarette smoking	Aromatic amine exposure	Real
Occupation	Aromatic amine and other chemical carcinogen exposure	Real
Schistosomiasis	Chronic inflammation	Real
Drugs		
Cyclophosphamide	Direct effect	Real
Phenacetin	Direct effect	Real
Family history	Genetic predisposition	Possible
Genetic polymorphisms (N-acetyltransferase 1, 2 [NAT1/2], glutathione S-transferase [GSTM1])	Inefficient detoxification of aromatic amines with subsequent increased excretion of potential carcinogens	Possible
Carcinogens in drinking water eg. chlorination byproducts, arsenic	Direct carcinogen exposure in the urine	Possible
Coffee	Carcinogenic metabolites in the urine	Controversial
Artificial sweeteners	Undefined in humans	Controversial

Question 2

What is the predominant histological type of bladder cancer seen in westernized countries and how do they generally present?

IN Western countries 90% of bladder cancer is transitional cell cancer of the bladder (TCCB). Squamous cell carcinoma is much less common in the West but its prevalence increases significantly in countries where infection with schistosoma is endemic. Adenocarcinoma accounts for a very small number of bladder cancer cases in Western countries, while non-epithelial bladder tumours are rare. Different histological types of bladder cancer occasionally coexist.

Symptoms: MOST patients present with **painless haematuria**. This can be macroscopic, leading to self-referral, or microscopic, detected on dipstick urinalysis. The likelihood of underlying pathology is greater with macroscopic than with microscopic haematuria. Therefore it is broadly accepted in urological practice that investigation of macroscopic haematuria should proceed swiftly from first presentation. Certain patients may also present with irritative lower urinary tract symptoms, including frequency, dysuria, hesitancy and alteration of urine flow, these symptoms tending to be more indicative of carcinoma in situ.
However, it is also notable that these symptoms can be caused by a wide variety of lower urinary tract pathology, most which is benign.
Advanced bladder cancer symptoms are related to the effects on other pelvic organs as well as including the constitutional symptoms of malignancy, weight loss, anorexia and lethargy.

Question 3

The urologist performed a cystoscopic examination which revealed a papillary lesion in the bladder. What investigative procedures are used in diagnosing bladder cancers?

CYSTOSCOPY of the lower urinary tract combined with imaging of the upper urinary tract is mandatory. While it is ideal to confirm the presence of red cells in the urine before proceeding with investigation; however, the nature of the malignant process in the bladder is such that bleeding is often intermittent and varying in intensity. The recommendation is to proceed with investigation when haematuria is reported in the patient's history or subsequent laboratory investigations.

Upper-tract imaging allows radiological visualisation of filling defects in the upper and lower urinary tracts, to assess for synchronous upper-tract lesions, and also aids in staging lower urinary tract cancers.

Cystoscopy of the lower urinary tract combined with imaging of the upper urinary tract (preferably a CT IV urography or standard IVU and renal tract ultrasound) is mandatory.
The initial request for diagnostic imaging should be concurrent with specialist referral, which minimises patient delay when they present to their designated urologist.

There is a continual search for an appropriate urine test that will be both sensitive and specific for bladder cancer. **Urinary cytology** is used by many and has an overall specificity of 96% but a sensitivity of only 49% when all grades and stages of bladder cancer are examined. However, it is much more reliable in the diagnosis of high-grade bladder cancer and carcinoma in situ. The holy grail of bladder cancer diagnosis is a sensitive and specific urinary marker. Many proposed markers have been tested but tend to be suboptimal in some respect.

Question 4

To plan appropriate therapy, establishing primary tumour stage and grade is vital. How is this done and how does it help in the management of the patient?

At present the best indicators for risk of recurrence and progression are primary disease stage and grade. Bladder tumours undergo primary endoscopic resection down to detrusor muscle, which includes sampling of this layer. Histopathological assessment of this resected tissue will establish whether the primary tumour is superficial and completely resected, or muscle invasive with the propensity for residual disease. The WHO grade will also be determined.

Tumour category is best assessed by histological examination of the resected specimen (pT category). The depth of penetration into or through the bladder wall is used as the criterion for the T component. A tumour is nominally regarded as superficial (pT1A) if it has penetrated no further than the basement membrane. Tumours that have transgressed the bladder wall to a greater degree are pT2-pT4, and their classification is based on a combination of histological and bimanual examination.

Accurate staging and histopathological diagnosis of bladder cancer is essential, as it remains the most valuable indicator for:

- **Appropriate selection of primary and adjuvant therapy**
- **Estimation of prognosis**
- **Assistance in evaluating the results of treatment**

- The exchange of accurate information among treatment centres

Grading of bladder cancer

The most widely used system for histopathological grading of bladder cancer is that introduced by the WHO in 1973. This classification divides TCCB as follows:

- GX — grade cannot be assessed.
- G1 — well differentiated.
- G2 — moderately differentiated.
- G3 — poorly differentiated.

Hence, after patients have been diagnosed and staged they will adopt an appropriate 'label' for their further care such as T1G3, T3aG3, etc.

Treatment depends on the site, size and histological grading of the tumour. At the initial endoscopic examination, the tumour is resected as far as possible, with deeper areas of resection being sent separately for histological examination to assess spread into the bladder muscle. Random biopsies of apparently normal bladder mucosa should also be obtained, as these may show changes of dysplasia or carcinoma in situ and provide useful prognostic information on the likelihood of recurrence.

Once a diagnosis of urothelial carcinoma has been made, regular lifelong follow-up is required; 50% of patients will develop further tumours.

Prognosis: 95% of those with pT1A tumours survive 5 years.